

Friend or fold?

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1 Setting

We are studying the model:

$$\dot{x}_i = x_i \left(r_i + (1 - \alpha) \sum_j A_{ij} x_j + \alpha \sum_j \sum_k B_{ijk} x_j x_k \right), \quad \alpha \in [0, 1], \quad (1)$$

without loss of generality, we can assume $B_{ijk} = B_{ikj}$, i.e., each tensor slice $B_{i..}$ is symmetric. We are interested in systems satisfying two conditions. First, we want $x^* = \mathbf{1}$ to be an equilibrium for all $\alpha \in [0, 1]$. This requires:

$$\sum_j A_{ij} = \sum_{j,k} B_{ijk} = -r_i$$

Second, we require $x^* = \mathbf{1}$ to be locally stable for any $\alpha \in [0, 1]$. The Jacobian at a positive equilibrium $x^* > \mathbf{0}$ is:

$$J(x^*, \alpha) = D(x^*)((1 - \alpha)A + 2\alpha\mathcal{B}(x^*)) = D(x^*)M(x^*, \alpha)$$

where the matrix $\mathcal{B}(x^*)$ is defined as:

$$\mathcal{B}(x^*)_{ij} = \sum_k B_{ijk} x_k^*$$

Write $\mathcal{B}(\mathbf{1}) = A + E$, then the first condition requires $A\mathbf{1} = \mathcal{B}(\mathbf{1})\mathbf{1} = -r$ and thus $E\mathbf{1} = \mathbf{0}$. We can write:

$$J(\mathbf{1}, \alpha) = M(\mathbf{1}, \alpha) = (1 - \alpha)A + 2\alpha\mathcal{B}(\mathbf{1}) = (1 + \alpha)A + 2\alpha E$$

and thus stability needs to be maintained along the segment connecting $A = M(\mathbf{1}, 0)$ to $2\mathcal{B}(\mathbf{1}) = M(\mathbf{1}, 1)$. This implies that both A and $\mathcal{B}(\mathbf{1})$ need to be Hurwitz. The converse does not hold in general: the Hurwitz matrices do not form a convex set, so stability of the two endpoints does not by itself guarantee stability along the whole segment.

2 Local bifurcations

We consider moving away from the equilibrium $x^* = \mathbf{1}$ in direction d :

$$x(\eta) = \mathbf{1} + \eta d, \quad \|d\| = 1, \quad \eta \geq 0. \quad (2)$$

and we want to know whether and when we will encounter a local bifurcation. As we step away from $x = \mathbf{1}$, one of two things can happen: (a) $x(\eta)$ leaves the positive cone (transcritical), or (b) $x(\eta) > 0$ loses stability.

First, let us analyze the case in which $x(\eta)$ touches the boundary of the positive cone. This happens whenever there exists a $d_i < 0$, at $\eta_i = -1/d_i$. Call $\eta_{\text{feas}} = 1/\max_i(-d_i)$ if any $d_i < 0$, and $\eta_{\text{feas}} = \infty$ otherwise; for d drawn uniformly on the sphere the latter case occurs with probability 2^{-n} , which is negligible for large n . Whenever $\eta \geq \eta_{\text{feas}}$ the equilibrium has left the interior of the positive cone. If stability was maintained up to this point, we have found a *transcritical bifurcation*: at $\eta = \eta_{\text{feas}}$ the interior branch meets the branch lying on the face $x_i = 0$ and the two exchange stability.

The loss of stability can further be separated into two main classes: (i) a Hopf bifurcation, in which a complex-conjugate pair of eigenvalues of $J(x(\eta), \alpha)$ crosses the imaginary axis into the right half-plane without making $J(x(\eta), \alpha)$ singular; (ii) a zero-eigenvalue bifurcation, in which there exists a critical η such that $\det J(x(\eta), \alpha) = 0$. The fold (codimension one) and its degenerations—cusp, Bogdanov–Takens, zero-Hopf—of higher codimension belong to this class. Since $\det J(x(\eta), \alpha) = (\prod_i x_i(\eta)) \det M(x(\eta), \alpha)$ and $\prod_i x_i(\eta) > 0$ in the interior, class (ii) occurs at an interior point only when $\det M(x(\eta), \alpha) = 0$.

For a fixed direction d and fixed α , the displacement η constitutes a single parameter. Generically, a one-parameter family encounters only codimension-one bifurcations, and the complete list of codimension-one *local* bifurcations available to a positive equilibrium of a Kolmogorov system is: boundary transcritical ($x_i \rightarrow 0$), fold ($\det M = 0$), and Hopf. Tracking these three therefore suffices to locate the loss of stable feasibility,

$$\eta^*(d, \alpha) = \min(\eta_{\text{feas}}, \eta_{\text{det}}, \eta_{\text{Hopf}}), \quad (3)$$

where $\eta_{\text{feas}} = 1/\max_i(-d_i)$ **does not depend on** α . Codimension-two points require a second defining condition and are therefore not encountered along a generic ray; they appear in the (η, α) plane as isolated organizing centers at which the corresponding codimension-one curves meet. We emphasize that this classification is local, in the sense that it exhausts the bifurcations detectable from $\text{spec } J(x(\eta))$; global codimension-one events (homoclinic orbits, saddle-node bifurcations of limit cycles, boundary crises) are equally generic in one parameter but leave the spectrum of the Jacobian unchanged, and may govern the fate of the system beyond a subcritical Hopf bifurcation.

In what follows we determine how α , the strength of HOIs, influences the values of η_{feas} and η_{det} .

3 Conditions for a fold bifurcation

We want to know whether and when the matrix $M(x(\eta), \alpha)$ becomes singular, as we vary η :

$$M(x(\eta), \alpha) = M(\mathbf{1}, \alpha) + 2\alpha\eta\mathcal{B}(d)$$

with $M(\mathbf{1}, \alpha)$ stable (and thus invertible) for all valid α . Then,

$$\det M(x(\eta), \alpha) = 0 \iff \det(I + 2\alpha\eta M(\mathbf{1}, \alpha)^{-1}\mathcal{B}(d)) = 0$$

If $Q(\alpha, d) = M(\mathbf{1}, \alpha)^{-1}\mathcal{B}(d)$ has no negative, real eigenvalue, then the determinant of $M(x(\eta), \alpha) \neq 0$ for all $\eta \geq 0$ and thus a fold bifurcation is impossible. We write this case as $\eta_{\det} = \infty$. At $\alpha = 0$, we have $M(x(\eta), 0) = M(\mathbf{1}, 0)$ and thus also in this case folds are impossible. Suppose now that $\alpha > 0$ and that $Q(\alpha, d)$ has a negative, real eigenvalue, $-\rho(\alpha, d)$: then the determinant will be zero at

$$\eta_{\det} = \frac{1}{2\alpha\rho(\alpha, d)}$$

If $Q(\alpha, d)$ has multiple real, negative eigenvalues, we consider the farthest from zero, which corresponds to the first crossing. While the function $\alpha\rho(\alpha, d)$ is in general very difficult to study, we can identify some cases in which the behavior can be determined mathematically.

4 Probability that $\eta_{\det} < \eta_{\text{feas}}$

Let

$$p(\alpha) = \Pr[\eta_{\det}(\alpha, d) < \eta_{\text{feas}}(d)]$$

be the probability, over directions d , that the $\eta_{\det}$ is reached before the equilibrium leaves the positive cone at η_{feas} . The question is how p varies with the strength of the higher-order interactions.

Lemma 1. *Suppose that for almost every direction d the map $\alpha \mapsto \eta_{\det}(\alpha, d)$ is non-increasing on $(0, 1]$. Then p is non-decreasing on $(0, 1]$.*

Proof. Let $S_\alpha = \{d : \eta_{\det}(\alpha, d) < \eta_{\text{feas}}(d)\}$. If $\alpha_1 < \alpha_2$ then $\eta_{\det}(\alpha_2, d) \leq \eta_{\det}(\alpha_1, d)$ for a.e. d , so $d \in S_{\alpha_1}$ implies $d \in S_{\alpha_2}$: the sets are nested increasing up to a null set. Monotonicity of the measure gives $p(\alpha_1) \leq p(\alpha_2)$. $\square$

Two features are important. First, the nesting holds *pointwise in d* , so the conclusion is distribution-free: it will hold for any distribution of d on the unit sphere. Second, it is only sufficient. Individual directions may fail to be monotone without destroying monotonicity of p , since the failures may cancel in the average. What this condition provides is a clean reduction: the entire question becomes the deterministic, linear-algebraic one of whether $\alpha\rho(\alpha, d)$ is non-decreasing in α for each fixed d . We analyze two cases in which $\alpha\rho(\alpha, d)$ is non-decreasing in α , and this result can be proven.

5 A special case

Suppose that $\sum_k B_{ijk} = A_{ij}$, and therefore $E = \mathbf{0}$. Then:

$$M(\mathbf{1}, \alpha) = (1 + \alpha)A$$

and its inverse:

$$M(\mathbf{1}, \alpha)^{-1} = \frac{1}{1 + \alpha}A^{-1}$$

Substituting, we find:

$$\det M(x(\eta), \alpha) = 0 \iff \det \left(I + \frac{2\alpha}{1+\alpha} \eta A^{-1} \mathcal{B}(d) \right) = 0$$

Suppose that $Q(d) = A^{-1} \mathcal{B}(d)$ (which depends **only** on d) has a negative, real eigenvalue, $-\rho(d)$ (take the one farthest from zero in case of multiple real, negative eigenvalues). Then,

$$\eta_{\det} = \frac{1+\alpha}{2\alpha \rho(d)}$$

and therefore,

$$\frac{\partial \eta_{\det}}{\partial \alpha} = -\frac{1}{2\alpha^2 \rho(d)} < 0$$

Because of this, map $\alpha \mapsto \eta_{\det}(\alpha, d)$ is non-increasing on $(0, 1]$, and $p(\alpha)$ is non-decreasing in α .

6 Symmetric A and B : a solvable case

Now consider the case in which $A_{ij} = A_{ji}$ (the matrix is symmetric) and, in addition to the slice symmetry $B_{ijk} = B_{ikj}$ assumed throughout, $B_{ijk} = B_{jik}$ (the two conditions generate invariance under all permutations of the indices, i.e., the tensor is symmetric). Then, we have that $\mathcal{B}(x(\eta))$ is a sum of symmetric matrices, and therefore is symmetric. Moreover, condition on the stability of $J(\mathbf{1}, \alpha)$ being stable implies that both A and $\mathcal{B}(\mathbf{1})$ are negative definite; conversely, because the negative definite matrices form a convex cone, negative definiteness of A and $\mathcal{B}(\mathbf{1})$ guarantees $M(\mathbf{1}, \alpha) \prec 0$ for every $\alpha \in [0, 1]$. The Jacobian at a positive x^* is the product of a diagonal and a symmetric matrix:

$$J(x^*, \alpha) = D(x^*) M(x^*, \alpha)$$

with $M(x^*, \alpha) = M^T(x^*, \alpha)$. Then $J(x^*, \alpha)$ is similar to a symmetric matrix and thus possesses only real eigenvalues:

$$D(x^*)^{-\frac{1}{2}} J(x^*, \alpha) D(x^*)^{\frac{1}{2}} = D(x^*)^{\frac{1}{2}} M(x^*, \alpha) D(x^*)^{\frac{1}{2}}$$

Because the Jacobian has only real eigenvalues for every $x^* > \mathbf{0}$ and α , Hopf bifurcations can be ruled out.

Because $J(x(\eta), \alpha)$ is congruent to the symmetric matrix $M(x(\eta), \alpha)$ via $D(x(\eta))^{1/2}$, it has real spectrum and the same inertia, so $x(\eta)$ is locally stable if and only if $M(x(\eta), \alpha) \prec 0$. For $\alpha \in (0, 1]$ write

$$P(\alpha) = -\frac{M(\mathbf{1}, \alpha)}{2\alpha} = \frac{1-\alpha}{2\alpha}(-A) + (-\mathcal{B}(\mathbf{1})) \succ 0, \quad \frac{M(x(\eta), \alpha)}{2\alpha} = -P(\alpha) + \eta \mathcal{B}(d).$$

Negative definiteness of the right-hand side requires $\eta y^\top \mathcal{B}(d) y < y^\top P(\alpha) y$ for every $y \neq \mathbf{0}$. For y with $y^\top \mathcal{B}(d) y \leq 0$ this is automatic; in particular, if $\mathcal{B}(d)$ is negative semi-definite no fold occurs and we set $\eta_{\det} = \infty$. Otherwise stability holds precisely for $\eta < \eta_{\det}$, where

$$\eta_{\det} = \min_{\substack{y \neq \mathbf{0} \\ y^\top \mathcal{B}(d) y > 0}} \frac{y^\top P(\alpha) y}{y^\top \mathcal{B}(d) y} = \min_{\substack{y \neq \mathbf{0} \\ y^\top \mathcal{B}(d) y > 0}} \frac{\frac{1-\alpha}{2\alpha} y^\top (-A) y + y^\top (-\mathcal{B}(\mathbf{1})) y}{y^\top \mathcal{B}(d) y}.$$

The ratio is scale invariant and tends to $+\infty$ as $y^\top \mathcal{B}(d)y \rightarrow 0^+$, so the minimum is attained. At $\eta = \eta_{\text{det}}$ the matrix $M(x(\eta), \alpha)$ is singular and the minimizer spans its kernel, so this is the fold. Monotonicity is now immediate: for each fixed y the denominator does not depend on α , while the numerator is strictly decreasing in α , since $(1 - \alpha)/2\alpha$ is strictly decreasing and $y^\top (-A)y > 0$. A minimum of strictly decreasing functions is strictly decreasing, so η_{det} is strictly decreasing in α on $(0, 1]$, from $\eta_{\text{det}} \rightarrow \infty$ as $\alpha \rightarrow 0^+$ to its smallest value at $\alpha = 1$.

7 Generic behavior

We have illustrated two cases in which $p(\alpha)$ is non-decreasing in α . Here we make some observations on the general behavior.

- We can have $p(\alpha) = 0$ for all α . Take the case of symmetric tensor, with $A \prec 0$ and $\mathcal{B}(\mathbf{1}) \prec 0$. Further, suppose that the slices $B_{..k} \preceq 0$: because $\mathcal{B}(x) = \sum_k x_k B_{..k}$, we have $\mathcal{B}(x) \preceq 0$ for every $x \geq \mathbf{0}$, and in fact $\mathcal{B}(x) \prec 0$ for every $x > \mathbf{0}$, since the kernel of $\mathcal{B}(x)$ is then the intersection of the kernels of the slices, which is the kernel of $\mathcal{B}(\mathbf{1}) = \sum_k B_{..k}$ and hence trivial. Therefore $M(x, \alpha) = (1 - \alpha)A + 2\alpha\mathcal{B}(x) \prec 0$ for every $x > \mathbf{0}$ and every $\alpha \in [0, 1]$, and folds are precluded everywhere in the positive cone.
- There is an upper bound on $p(\alpha)$. For example, in the symmetric case, a fold requires $\mathcal{B}(d) \not\preceq 0$, and therefore $p(\alpha) \leq 1 - \Pr[\mathcal{B}(d) \preceq 0]$. This bound is nontrivial: $\mathcal{B}(\mathbf{1}) \prec 0$ and $\mathcal{B}$ is linear, so $\mathcal{B}(d) \prec 0$ on an open neighborhood of $d_0 = \mathbf{1}/\sqrt{n}$, and hence $\Pr[\mathcal{B}(d) \preceq 0] > 0$.
- $p(1) < 1$ also without assuming symmetry. At $\alpha = 1$ a fold along d requires $Q(1, d) = (2\mathcal{B}(\mathbf{1}))^{-1}\mathcal{B}(d)$ to have a negative, real eigenvalue. Take $d_0 = \mathbf{1}/\sqrt{n}$: by linearity $\mathcal{B}(d_0) = \mathcal{B}(\mathbf{1})/\sqrt{n}$, so $Q(1, d_0) = I/(2\sqrt{n})$, whose spectrum is the single positive value $1/(2\sqrt{n})$. Since the eigenvalues of $Q(1, d)$ are continuous functions of d alone, they remain in a small neighborhood of $1/(2\sqrt{n})$, and in particular none of them is real and negative, for an open neighborhood of d_0 : no folds occur along those directions.